

Mata Kuliah : Praktikum Algoritma dan Struktur Data
Program Studi : D4 – Sistem Informasi Bisnis
Semester : 2

Kelas : SIB 1C
NIM : 254107060025
Nama : Ananda Vesakhagotama Suparman
CM : 2
Link GitHub : <https://github.com/vesakhaa/CM2>

1. Pembeli04.java

```
1 public class Pembeli04 {  
2     int noAntrian;  
3     String namaPembeli, noHP;  
4     Pembeli04 prev, next;  
5  
6     public Pembeli04(int noAntrian, String namaPembeli, String noHP) {  
7         this.noAntrian = noAntrian;  
8         this.namaPembeli = namaPembeli;  
9         this.noHP = noHP;  
10        this.prev = null;  
11        this.next = null;  
12    }  
13 }
```

2. Pesanan04.java

```
1 public class Pesanan04 {  
2     int kodePesanan, harga;  
3     String namaPesanan;  
4     Pesanan04 prev, next;  
5  
6     public Pesanan04(int kodePesanan, String namaPesanan, int harga) {  
7         this.kodePesanan = kodePesanan;  
8         this.namaPesanan = namaPesanan;  
9         this.harga = harga;  
10        this.prev = null;  
11        this.next = null;  
12    }  
13 }  
14 }
```

3. DLLAntrian04.java

```
1 public class DLLAntrian04 {
2     Pembeli04 head, tail;
3     int CurrentAntrian = 1;
4
5     public void enqueue(String nama, String noHP) {
6         Pembeli04 newNode = new Pembeli04(CurrentAntrian++, nama, noHP);
7         if (head == null) {
8             head = tail = newNode;
9         } else {
10             tail.next = newNode;
11             newNode.prev = tail;
12             tail = newNode;
13         }
14         System.out.println("Antrian berhasil ditambahkan dengan nomor: " + newNode.noAntrian);
15     }
16
17     public void printQueue() {
18         if (head == null) {
19             System.out.println("Antrian Kosong");
20             return;
21         }
22         System.out.println("=====");
23         System.out.println("Daftar Antrian Pembeli");
24         System.out.println("=====");
25         System.out.printf("%-15s %-15s %-15s\n", "No Antrian", "Nama", "No HP");
26         Pembeli04 current = head;
27         while (current != null) {
28             System.out.printf("%-15d %-15s %-15s\n", current.noAntrian, current.namaPembeli,
29 current.noHP);
30             current = current.next;
31         }
32     }
33
34     public Pembeli04 dequeue() {
35         if (head == null) {
36             return null;
37         }
38         Pembeli04 removedNode = head;
39         head = head.next;
40         if (head != null) {
41             head.prev = null;
42         } else {
43             tail = null;
44         }
45         return removedNode;
46     }
47 }
```

4. DLLPesanan04.java

```
1 public class DLLPesanan04 {
2     Pesanan04 head, tail;
3
4     public void addPesanan(int kode, String nama, int harga) {
5         Pesanan04 newNode = new Pesanan04(kode, nama, harga);
6         if (head == null) {
7             head = tail = newNode;
8         } else {
9             tail.next = newNode;
10            newNode.prev = tail;
11            tail = newNode;
12        }
13    }
14
15    public void sortPesananByNama() {
16        if (head == null || head.next == null) return;
17
18        boolean swapped;
19        Pesanan04 current;
20
21        do {
22            swapped = false;
23            current = head;
24
25            while (current.next != null) {
26                if (current.namaPesanan.compareToIgnoreCase(current.next.namaPesanan) > 0) {
27                    int tempKode = current.kodePesanan;
28                    String tempNama = current.namaPesanan;
29                    int tempHarga = current.harga;
30
31                    current.kodePesanan = current.next.kodePesanan;
32                    current.namaPesanan = current.next.namaPesanan;
33                    current.harga = current.next.harga;
34
35                    current.next.kodePesanan = tempKode;
36                    current.next.namaPesanan = tempNama;
37                    current.next.harga = tempHarga;
38
39                    swapped = true;
40                }
41                current = current.next;
42            }
43        } while (swapped);
44    }
45
46    public void printPesanan() {
47        if (head == null) {
48            System.out.println("Belum ada data pesanan.");
49            return;
50        }
51
52        sortPesananByNama();
53
54        System.out.println("=====");
55        System.out.println("LAPORAN PESANAN");
56        System.out.println("=====");
57        System.out.printf("%-15s %-20s %-15s\n", "Kode Pesanan", "Nama Pesanan", "Harga");
58        Pesanan04 current = head;
59        while (current != null) {
60            System.out.printf("%-15d %-20s %-15d\n", current.kodePesanan, current.namaPesanan,
61                current.harga);
62            current = current.next;
63        }
64    }
```

5. Main04.java

```
1 import java.util.Scanner;
2
3 public class Main04 {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         DLLAntrian04 antrian = new DLLAntrian04();
7         DLLPesanan04 pesanan = new DLLPesanan04();
8
9         int menu;
10        do {
11            System.out.println("=====");
12            System.out.println("SISTEM ANTRIAN ROYAL DELISH");
13            System.out.println("=====");
14            System.out.println("1. Tambah Antrian");
15            System.out.println("2. Cetak Antrian");
16            System.out.println("3. Hapus Antrian dan Pesan");
17            System.out.println("4. Laporan Pesanan");
18            System.out.println("0. Keluar");
19            System.out.print("Pilih menu : ");
20            menu = sc.nextInt();
21            sc.nextLine();
22
23            switch (menu) {
24                case 1:
25                    System.out.print("Nama Pembeli : ");
26                    String nama = sc.nextLine();
27                    System.out.print("No HP      : ");
28                    String hp = sc.nextLine();
29                    antrian.enqueue(nama, hp);
30                    break;
31                case 2:
32                    antrian.printQueue();
33                    break;
34                case 3:
35                    Pembeli04 pembeliDipanggil = antrian.dequeue();
36                    if (pembeliDipanggil != null) {
37                        System.out.print("Kode Pesanan : ");
38                        int kode = sc.nextInt();
39                        sc.nextLine();
40                        System.out.print("Nama Pesanan : ");
41                        String namaPesanan = sc.nextLine();
42                        System.out.print("Harga      : ");
43                        int harga = sc.nextInt();
44                        sc.nextLine();
45
46                        pesanan.addPesanan(kode, namaPesanan, harga);
47                        System.out.println(pembeliDipanggil.namaPembeli + " telah memesan " + namaPesanan);
48                    } else {
49                        System.out.println("Antrian saat ini kosong!");
50                    }
51                    break;
52                case 4:
53                    pesanan.printPesanan();
54                    break;
55                case 0:
56                    System.out.println("Keluar dari sistem...");
57                    break;
58                default:
59                    System.out.println("Pilihan menu tidak valid!");
60            }
61            System.out.println();
62        } while (menu != 0);
63
64        sc.close();
65    }
66 }
```

Hasil:

```
vesakhaaz@vesakhaaz:~/run/media/vesakhaaz/Linux only/ASD/OM25
_d65fe5ba/bin$ ./Main04

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 1
Nama Pembeli : Farel
No HP      : 123456
Antrian berhasil ditambahkan dengan nomor: 1

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 1
Nama Pembeli : Rifki
No HP      : 654321
Antrian berhasil ditambahkan dengan nomor: 2

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 1
Nama Pembeli : Reza
No HP      : 123123
Antrian berhasil ditambahkan dengan nomor: 3

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 2

=====
Daftar Antrian Pembeli
=====
No Antrian   Nama      No HP
1            Farel    123456
2            Rifki    654321
3            Reza     123123

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 3
Kode Pesanan : 01
Nama Pesanan : Pizza
Harga      : 25000
Farel telah memesan Pizza

=====
```

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
```

- ```
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 2
```

```
=====
Daftar Antrian Pembeli
=====
```

```
No Antrian Nama No HP
2 Rifki 654321
3 Reza 123123
```

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
```

- ```
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : 4
```

```
=====
LAPORAN PESANAN
=====
```

```
Kode Pesanan  Nama Pesanan  Harga
1             Pizza      25000
```

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
```

- ```
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu : █
```